

TAKUDZWA VUMA

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Education

Bachelor of Science in Computer Science

May 2027

Talladega College

GPA: 3.98

Professional Affiliations: Member of ColorStack, National Society of Black Engineers (NSBE), CodePath, ACM, and Thurgood Marshall College Fund (TMCf)

Technical Skills

Programming & Scripting: Python, HTML, CSS, Bash, PowerShell, SQL, JSON, Interactive AI Tooling

Security & Frameworks: NIST CSF, MITRE ATT&CK, Incident Response, Digital Forensics, Threat Detection, Intrusion Detection (Snort, Suricata), Vulnerability Assessment

Cloud & Identity (IAM): Microsoft Azure, AWS, Entra ID (Azure AD), Active Directory, Okta, Azure RBAC, Multi-Factor Authentication (MFA), Least Privilege, Account Lifecycle Management

Networking & Security Controls: Fortinet Firewalls, iptables, Cisco Switching, Wireshark (Packet Analysis), TCP/IP, DNS/DHCP, Cryptography/Encryption, Perimeter Protection

Operating Systems & Platforms: Linux (Kali, Ubuntu), Windows, macOS, Splunk, ServiceNow, hping3.

Experience

Undergraduate Research Assistant

August 2025 – Present

Talladega College

Talladega, AL

- Simulated DoS attack vectors (MITRE ATT&CK T1498/T1499) using hping3 across a 3-VM lab to stress-test perimeter defenses.
- Tuned Snort and Suricata IDS rules across 5 scenarios, boosting detection accuracy to 95% and cutting false positives to 5%.
- Investigated raw packet captures in Wireshark to isolate threat patterns, triggering automated iptables containment rules.
- Authored a technical benchmarking paper mapping detection metrics directly to NIST CSF (Detect & Respond) functions.

Information Technology Support Specialist

Jan 2025 – May 2025

Talladega College

Talladega, AL

- Administered IAM across Active Directory, Entra ID, and Okta for 300 users, executing 60+ lifecycle actions while enforcing RBAC and MFA compliance.
- Triaged MFA failures, account lockouts, and unauthorized access attempts across 300 identities, documenting audit trails to secure access.
- Hardened 40 workstations, 10 printers, and wireless infrastructure across 5 facilities, applying baseline endpoint controls.
- Authored 10+ internal KB articles on access provisioning, driving a 20% reduction in recurring identity tickets.

Technical Support Engineer

August 2021 – July 2023

Telone Zimbabwe

Harare, Zimbabwe

- Remediated 5–10 critical network faults weekly across fiber, ADSL, and VoIP for 60 clients, maintaining a 90% Service Level Agreements ticket closure rate.
- Correlated CPE telemetry, traffic logs, and routing metrics with a 3-person team to isolate infrastructure faults from edge failures.
- Automated remote patch management and configuration for 20 CPE routers weekly via TR-069/ACS protocols.
- Deployed Fortinet firewalls, Cisco switches, and edge routers during field operations, validating packet delivery before service activation.

Technical Projects

Azure IAM & Cloud Hardening Lab | Azure, Entra ID, AD DS, PowerShell, Interactive AI Tooling

January 2021

- Architected a hybrid cloud lab with 5 Azure VMs and AD DS using PowerShell and interactive AI tooling to automate provisioning
- Audited cloud permissions across 25 user accounts, enforcing Azure RBAC to cut excessive standing privileges by 40%.

Leadership / Extracurricular

Technical Cybersecurity Lead

August 2025 – Present

Google Developer Student Club (GDSC)

Talladega College

- Lead a group of 13 students in conducting hands-on cybersecurity workshops covering threat detection, network defense, and cloud IAM fundamentals.
- Tutor 13 peers on industry frameworks (NIST CSF, MITRE ATT&CK) and practical security tools (Wireshark, Linux, Snort) to accelerate skill development.